

1.8. Unnormalized Form (UNF)

1.8.1. Description

The raw Flight data collected before any normalization is applied. The Route column stores two pieces of information (origin and destination) in a single composite field. The Airline column stores a name directly instead of a reference, creating redundancy.

1.8.2. Identified Problems

- Violation: Route column stores two values in one field: 'Addis Ababa → Nairobi' — not atomic.
- Violation: AirlineName stored directly — same airline name repeated across multiple flight rows.
- Violation: No separation of concerns — fl

1.8.3. UNF Flight Table (4 Records)

FlightID	FlightNo	origin -> destniction	DateTime	Airline
1	ET101	Addis Ababa → Nairobi	2026-05-01 08:00	Ethiopian Airlines
2	KQ202	Nairobi → Addis Ababa	2026-05-02 14:00	Kenya Airways
3	EK711	Addis Ababa → Dubai	2026-05-03 22:00	Emirates
4	MS961	Nairobi → Cairo	2026-05-04 10:30	EgyptAir

1.9.1NF Flight Table (4 Records)

All values are now atomic. Primary key (FlightID) uniquely identifies each row.

FlightID (PK)	FlightNo	Origin	Destination	DateTime	AirlineID (FK)	AirlineName
1	ET101	Addis Ababa	Nairobi	2026-05-01 08:00	1	Ethiopian Airlines
2	KQ202	Nairobi	Addis Ababa	2026-05-02 14:00	2	Kenya Airways
3	EK711	Addis Ababa	Dubai	2026-05-03 22:00	3	Emirates
4	MS961	Nairobi	Cairo	2026-05-04 10:30	4	EgyptAir

2.Second Normal Form (2NF)

2.1. Rule

Must be in 1NF, and every non-key attribute must be fully functionally dependent on the entire primary key. No partial dependencies are allowed.

2.2.Changes Applied

Violation: AirlineName depends only on AirlineID, not on FlightID. This is a partial dependency causing update anomalies — if an airline renames itself, every flight row must be updated.

Fix: Extracted Airline into its own table with AirlineID (PK) and AirlineName. Flight table retains AirlineID as a Foreign Key. Now all non-key attributes in Flight depend fully on FlightID.

2.3. Flight Table — After 2NF

FlightID (PK)	FlightNo	Origin	Destination	DateTime	AirlineID (FK)
1	ET101	Addis Ababa	Nairobi	2026-05-01 08:00	1
2	KQ202	Nairobi	Addis Ababa	2026-05-02 14:00	2
3	EK711	Addis Ababa	Dubai	2026-05-03 22:00	3
4	MS961	Nairobi	Cairo	2026-05-04 10:30	4

2.4. Airline Table (Extracted)

AirlineID (PK)	AirlineName
1	Ethiopian Airlines
2	Kenya Airways
3	Emirates
4	EgyptAir

Foreign Key Relationship

Flight.AirlineID: → references Airline(AirlineID)

3. Third Normal Form (3NF)

3.1. Rule

Must be in 2NF, and no non-key attribute may depend on another non-key attribute (no transitive dependencies). Every non-key attribute must depend directly on the primary key only.

3.2. Changes Applied

Violation: Origin and Destination store city names directly. If airport details (country, code, timezone) were needed, they would depend on the city name — not on FlightID — creating a transitive dependency.

Fix: Introduced an Airport table. Origin and Destination are replaced by OriginID (FK) and DestinationID (FK) referencing Airport(AirportID). Now all non-key attributes in Flight depend solely on FlightID.

3.3. Airport Table (New)

AirportID (PK)	CityName	Country
1	Addis Ababa	Ethiopia
2	Nairobi	Kenya
3	Dubai	UAE
4	Cairo	Egypt

3.4. Airline Table (Unchanged)

AirlineID (PK)	AirlineName
1	Ethiopian Airlines
2	Kenya Airways
3	Emirates
4	EgyptAir

3.5. Flight Table — Final 3NF

All city name strings replaced with foreign key references to the Airport table.

FlightID (PK)	FlightNo	OriginID (FK)	DestID (FK)	DateTime	AirlineID (FK)
1	ET101	1	2	2026-05-01 08:00	1
2	KQ202	2	1	2026-05-02 14:00	2
3	EK711	1	3	2026-05-03 22:00	3
4	MS961	2	4	2026-05-04 10:30	4